CUSTOMER PURCHASE BEHAVIOR AND SALES PERFORMANCE ANALYSIS

1. Project Overview

This project focuses on analyzing customer purchase transactions to identify sales trends, customer buying behavior, product performance, payment preferences, and revenue generation patterns. The dataset contains purchase records across multiple product categories, including Electronics, Furniture, and Office Supplies, along with details such as quantity purchased, unit price, customer ratings, payment methods, and total sales value.

Objective

- Calculate total sales revenue.
- Identify top-selling products.
- Analyze category-wise sales performance.
- Study customer purchasing behavior.
- Evaluate payment method preferences.
- Monitor monthly sales trends.
- Analyze customer review ratings.
- Create an interactive Power BI dashboard for business insights.

2. Data Sources

- **Source Description:** https://trumpexcel.com/excel-sample-data/
- **Domain**: Retail / Sales Analytics etc

3. Problem Statement

- Identify total sales generated by the business.
- Find the top-selling products based on revenue.
- Analyze which product category performs best.
- Understand customer purchasing behavior.
- Track monthly sales trends and patterns.
- Identify the most preferred payment method.
- Analyze customer ratings and satisfaction levels.
- Find high-value customers based on purchase amount.
- Compare quantity sold across different product categories.
- Develop an interactive dashboard for business decision-making**.**

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Customer ID	Text/String	Unique identifier for each customer
Customer Name	Text	Name of the customer
Product	Text	Name of the purchased product
Product Category	Text	Category to which the product belongs
Purchase Date	Date	Date on which the product was purchased
Quantity	Whole Number	Number of units purchased
Unit Price	Decimal Number	Price per unit of the product

Total Price	Decimal Number	Total purchase amount
Payment Method	Text	Method used to make payment
Review Rating	Whole Number	Rating provided by the customer

5. Tools & Technologies

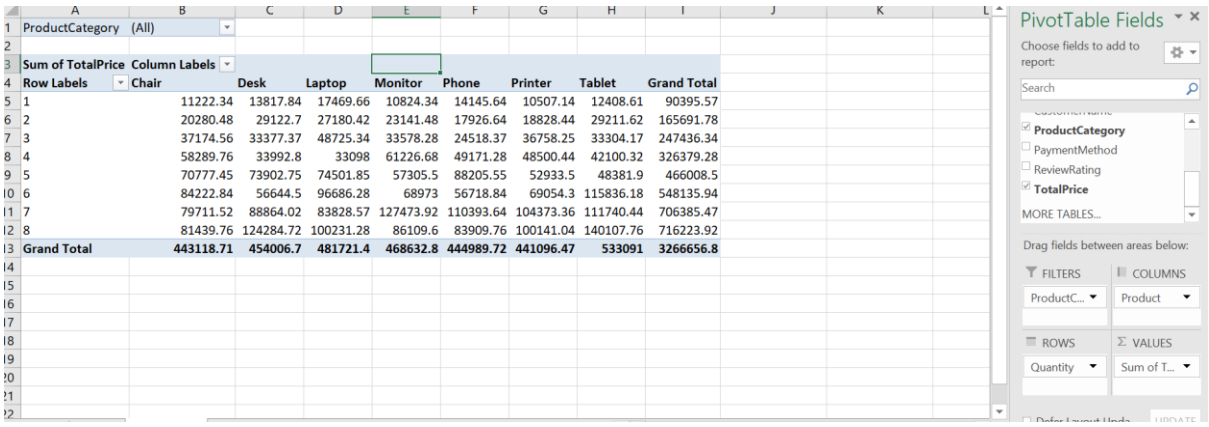
- **Microsoft Excel:** Reviewing data, cleaning records, checking formats, and preparing the source table.
- **Power BI Desktop:** Importing data, creating calculated columns and measures, building visuals, and designing the report.
- **Power Query:** Checking column names and data types and applying transformation steps during import.
- **DAX (Data Analysis Expressions):** Creating calculations such as total sales, customer count, average rating, and repeat-customer rate.

6. Data Pre-Processing:

Tasks Performed:

- Reviewed the column names and confirmed that the dataset contains customer, product, date, quantity, price, payment, and rating information.
- Checked data types: CustomerID and text attributes as Text; PurchaseDate as Date; Quantity as Whole Number; UnitPrice and TotalPrice as Decimal Number; ReviewRating as Whole Number.
- Checked for blank values, duplicate records, inconsistent text, and invalid date or numeric formats.
- Standardized the data format and ensured that the data was suitable for analysis.
- Kept the data in a single purchase table because the source workbook contains one sheet.
- Loaded the prepared data into Power BI.

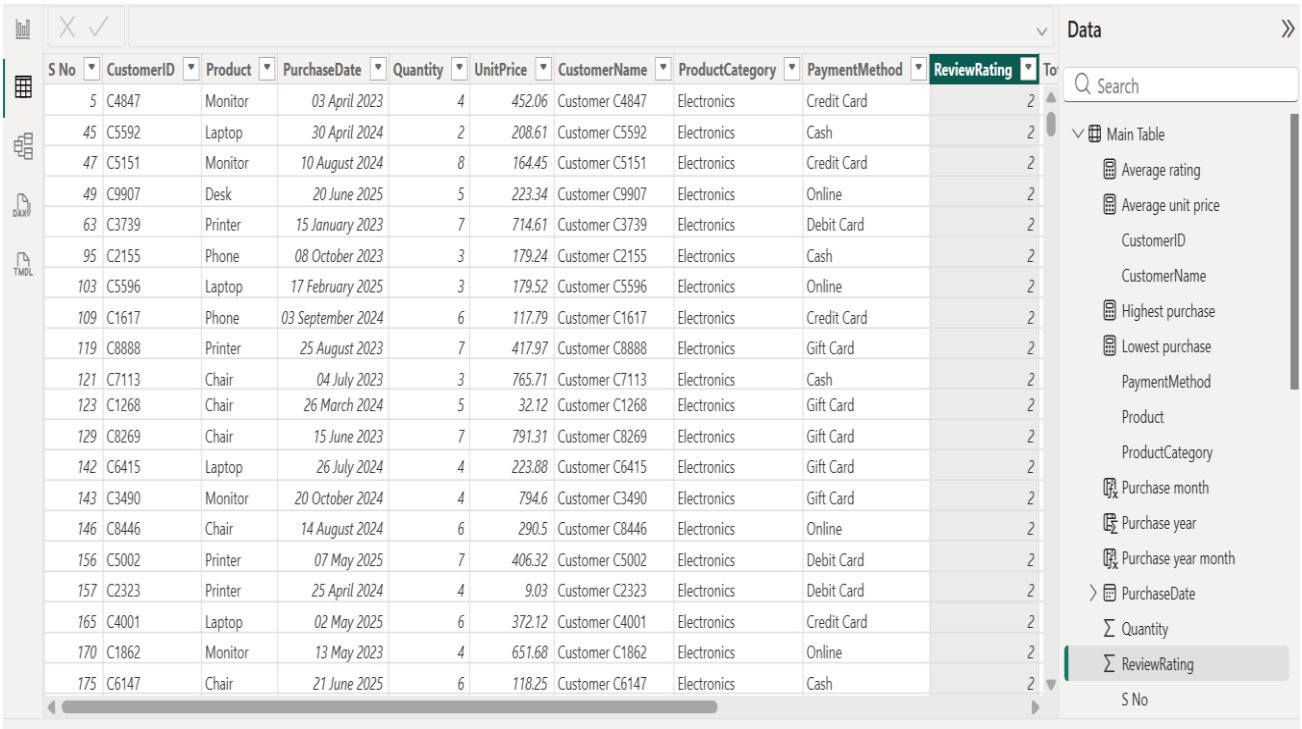
S No	CustomerI	Produc	PurchaseDat	Quantit	UnitPric	CustomerNam	ProductCategor	PaymentMetho	ReviewRatin	TotalPric
1	C5361	Phone	05-03-2024	8	619	Customer C5361	Office Supplies	Cash	1	4951
2	C6231	Laptop	21-06-2025	7	366	Customer C6231	Electronics	Debit Card	3	2564
3	C7704	Chair	25-06-2023	5	635	Customer C7704	Office Supplies	Credit Card	4	3173
4	C2923	Printer	30-09-2023	3	509	Customer C2923	Office Supplies	Gift Card	1	1526
5	C4847	Monitor	03-04-2023	4	452	Customer C4847	Electronics	Credit Card	2	1808
6	C7317	Chair	11-04-2023	3	210	Customer C7317	Office Supplies	Gift Card	1	629
7	C5076	Desk	19-06-2024	3	89	Customer C5076	Office Supplies	Debit Card	3	266
8	C5181	Chair	16-08-2023	8	274	Customer C5181	Furniture	Credit Card	2	2191
9	C9564	Monitor	11-02-2023	6	719	Customer C9564	Electronics	Gift Card	3	4313
10	C7521	Tablet	14-06-2025	7	100	Customer C7521	Office Supplies	Debit Card	4	703
11	C7713	Desk	24-09-2023	2	453	Customer C7713	Electronics	Credit Card	4	906
12	C3269	Desk	25-04-2025	7	528	Customer C3269	Furniture	Debit Card	4	3698
13	C7058	Laptop	28-09-2024	1	540	Customer C7058	Furniture	Cash	3	540
14	C8718	Phone	26-01-2025	2	797	Customer C8718	Electronics	Credit Card	3	1595
15	C2966	Phone	16-11-2023	8	284	Customer C2966	Electronics	Cash	1	2272
16	C7436	Laptop	20-11-2024	8	570	Customer C7436	Electronics	Debit Card	5	4556
17	C8827	Phone	06-09-2023	5	468	Customer C8827	Furniture	Online	2	2338
18	C1067	Monitor	01-12-2024	5	601	Customer C1067	Electronics	Debit Card	4	3003
19	C3719	Tablet	12-10-2024	7	7	Customer C3719	Office Supplies	Gift Card	5	48
20	C8138	Monitor	29-09-2023	3	195	Customer C8138	Furniture	Cash	5	584
21	C7824	Chair	15-06-2024	2	559	Customer C7824	Office Supplies	Online	1	1118



ProductCategory	(All)							
Sum of TotalPrice	Column Labels							
Row Labels	Chair	Desk	Laptop	Monitor	Phone	Printer	Tablet	Grand Total
1	11222.34	13817.84	17469.66	10824.34	14145.64	10507.14	12408.61	90395.57
2	20280.48	29122.7	27180.42	23141.48	17926.64	18828.44	29211.62	165691.78
3	37174.56	33377.37	48725.34	33578.28	24518.37	36758.25	33304.17	247436.34
4	58289.76	33992.8	33098	61226.68	49171.28	48500.44	42100.32	326379.28
5	70777.45	73902.75	74501.85	57305.5	88205.55	52933.5	48381.9	466008.5
6	84222.84	56644.5	96686.28	68973	56718.84	69054.3	115836.18	548135.94
7	79711.52	88864.02	83828.57	127473.92	110393.64	104373.36	111740.44	706385.47
8	81439.76	124284.72	100231.28	86109.6	83909.76	100141.04	140107.76	716223.92
Grand Total	443118.71	454006.7	481721.4	468632.8	444989.72	441096.47	533091	3266656.8

7. Data Modelling and DAX (Power BI)

Data Model: The project uses one main table (Customer_Purchases). Since the data is contained in one table, no table relationships are required for the current dashboard.



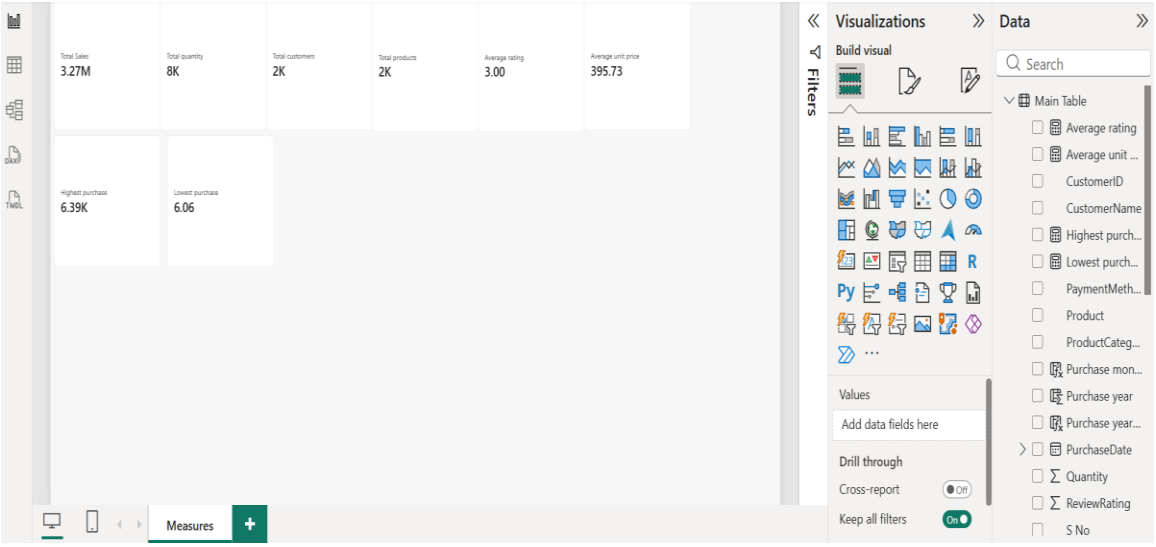
S No	CustomerID	Product	PurchaseDate	Quantity	UnitPrice	CustomerName	ProductCategory	PaymentMethod	ReviewRating	To
5	C4847	Monitor	03 April 2023	4	452.06	Customer C4847	Electronics	Credit Card	2	
45	C5592	Laptop	30 April 2024	2	208.61	Customer C5592	Electronics	Cash	2	
47	C5151	Monitor	10 August 2024	8	164.45	Customer C5151	Electronics	Credit Card	2	
49	C9907	Desk	20 June 2025	5	223.34	Customer C9907	Electronics	Online	2	
63	C3739	Printer	15 January 2023	7	714.61	Customer C3739	Electronics	Debit Card	2	
95	C2155	Phone	08 October 2023	3	179.24	Customer C2155	Electronics	Cash	2	
103	C5596	Laptop	17 February 2025	3	179.52	Customer C5596	Electronics	Online	2	
109	C1617	Phone	03 September 2024	6	117.79	Customer C1617	Electronics	Credit Card	2	
119	C8888	Printer	25 August 2023	7	417.97	Customer C8888	Electronics	Gift Card	2	
121	C7113	Chair	04 July 2023	3	765.71	Customer C7113	Electronics	Cash	2	
123	C1268	Chair	26 March 2024	5	32.12	Customer C1268	Electronics	Gift Card	2	
129	C8269	Chair	15 June 2023	7	791.31	Customer C8269	Electronics	Gift Card	2	
142	C6415	Laptop	26 July 2024	4	223.88	Customer C6415	Electronics	Gift Card	2	
143	C3490	Monitor	20 October 2024	4	794.6	Customer C3490	Electronics	Gift Card	2	
146	C8446	Chair	14 August 2024	6	290.5	Customer C8446	Electronics	Online	2	
156	C5002	Printer	07 May 2025	7	406.32	Customer C5002	Electronics	Debit Card	2	
157	C2323	Printer	25 April 2024	4	9.03	Customer C2323	Electronics	Debit Card	2	
165	C4001	Laptop	02 May 2025	6	372.12	Customer C4001	Electronics	Credit Card	2	
170	C1862	Monitor	13 May 2023	4	651.68	Customer C1862	Electronics	Online	2	
175	C6147	Chair	21 June 2025	6	118.25	Customer C6147	Electronics	Cash	2	

Calculated columns:

- **Purchase Year =** YEAR(Customer_Purchases[PurchaseDate])
- **Purchase Month =** FORMAT(Customer_Purchases[PurchaseDate], "MMM")
- **Month Number =** MONTH(Customer_Purchases[PurchaseDate])
- **Purchase Year Month =** FORMAT(Customer_Purchases[PurchaseDate], "YYYY-MM")

DAX Measures:

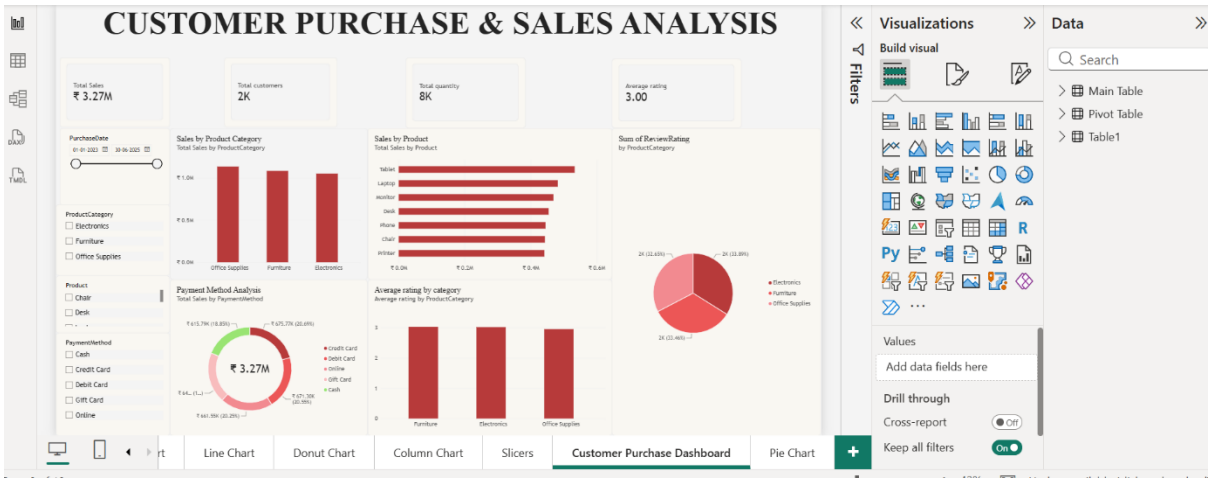
- Total Sales- SUM('Main Table'[TotalPrice])
- Total Quantity- SUM('Main Table'[Quantity])
- Total Customers- COUNT('Main Table'[CustomerID])
- Total Products- COUNT('Main Table'[Product])
- Average Rating- AVERAGE('Main Table'[ReviewRating])
- Average Unit Price- AVERAGE('Main Table'[UnitPrice])
- Highest Purchase- MAX('Main Table'[TotalPrice])
- Lowest Purchase- MIN('Main Table'[TotalPrice])



8. Analysis and Visualizations (Power BI)

Dashboard Features:

- Card – Total Sales
- Card – Total Customers
- Card – Total Quantity
- Card – Average Rating
- Clustered column chart – Sales by Product Category
- Bar chart – Top 5 Products by Total Sales
- Line chart – Monthly Sales Trend
- Donut chart – Sales by Payment Method
- Bar chart – Top Customers by Sales
- Column chart – Average Rating by Product Category
- Slicers – Purchase Date, Product Category, Product, Payment Method, and Customer Name



9. Insights:

Key Findings:

- Purchase records: 1,800
- Distinct customers: 1,642
- Total sales: 3,266,656.80
- Total quantity: 8,260
- Average review rating: approximately 3.00 out of 5
- Office Supplies has the highest category sales (1,133,462.57), followed by Furniture (1,083,282.25) and Electronics (1,049,911.98).
- Tablet is the highest-sales product label (533,091.00).

Descriptive Analysis:

The report summarises sales, customer counts, quantities, category performance, product performance, monthly sales, payment methods, and review ratings.

Diagnostic Analysis:

The dashboard compares product categories, products, payment methods, and customer ratings.

Key observations:

- Office Supplies generates more sales than Furniture and Electronics.
- Tablet generates the highest sales value
- Sales contributions differ across payment methods.

Predictive Analysis:

This data covers only January to June only in the year of 2025. We assuming the second half performs like the first half. This is an assumption-based not a formal forecast.

Prescriptive Analysis:

- Maintain availability of products that contribute strongly to sales.
- Investigate lower-sales categories and products before making pricing or marketing changes.
- Review customer feedback and identify products/categories that may need service or quality improvements.
- Test customer retention activities and track repeat-customer measures over time.
- Collect additional data such as costs, inventory, promotions, and customer feedback comments for stronger business recommendations.

10. Conclusion:

This project demonstrates an end-to-end retail purchase analysis workflow using Excel and Power BI. The source data was reviewed and prepared, DAX measures were created to calculate key metrics, and an interactive dashboard was designed to explore sales, products, customers, payment methods, and ratings. The dashboard makes the data easier to understand and supports evidence-based discussion.